

THE RELATIONSHIP BETWEEN WORKING MEMORY FOR SERIAL ORDER AND  
NUMERICAL DEVELOPMENT: A LONGITUDINAL STUDY

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**Abstract**

Despite numerous studies, the link between verbal working memory (WM) and calculation abilities remains poorly understood. The present longitudinal study focuses specifically on the role of serial order retention capacities, based on recent findings suggesting a link between ordinal processing in verbal WM and numerical processing tasks. Children were tested when they were in third year kindergarten (T1), first grade (T2) and second grade of primary school (T3), with WM tasks maximizing retention of serial order or item information, as well as with numerical judgment and calculation tasks. We observed that order WM measures at T1 provided a robust predictor of calculation abilities at T2 and T3. Numerical ordinal and magnitude judgment abilities were also associated with calculation abilities and this, independently of order WM abilities. This study highlights the important role of WM for order in early calculation acquisition, in addition to numerical ordinal and magnitude representations, and provides new perspectives for our understanding of the link between verbal WM and numerical abilities.

164 words

**Keywords:** Short term memory; Working memory; Order processing; Numerical development; Ordinal processing.

## **Introduction**

Many studies have explored the role of working memory (WM) in numerical cognition and its development. These studies have led to conflicting results, especially with respect to the intervention of verbal WM. Most of these studies were based on the architecture of the WM model by Baddeley and Hitch (1974) distinguishing between three components: A processing component, the central executive, and two passive storage systems, the visuospatial sketchpad for visuo-spatial information and the phonological loop for verbal information. A reliable association has been observed between the central executive component of WM and numerical capacities such as arithmetic problem solving (Bull & Scerif, 2001; De Smedt et al., 2009; Gathercole & Pickering, 2000a; Geary, Hamson, & Hoard, 2000; Geary, Hoard, Byrd-Craven, Nugent, & Numtee, 2007; Noël, 2009; Passolunghi, Vercelloni, & Schadee, 2007; Swanson & Beebe-Frankenberger, 2004; Swanson & Kim, 2007). On the other hand, the relationship between the passive storage systems and numerical abilities remains controversial, especially with respect to the verbal short-term storage system. While visuo-spatial short-term retention abilities have been related to numerical abilities (Holmes, Adams, & Hamilton, 2008; Jarvis & Gathercole, 2003), evidence for a link between verbal short-term retention abilities and numerical abilities is much less consistent (Hecht, Torgesen, Wagner & Rashotte, 2001; Gathercole & Pickering, 2000b; Holmes & Adams, 2006; Bull & Johnston, 1997; Geary et al., 2000; De Smedt et al., 2009; Noël, Seron and Tovarelli, 2004). The aim of the present study is to gain a clearer understanding of the relationship between verbal short-term retention abilities and the development of numerical abilities by focusing on a critical but neglected dimension of verbal WM, the processing and storage of serial order information.

In most studies on verbal WM and numerical development, retention of serial order information and item information are confounded, where item information refers to the

identity of the items of a memory list, and order information refers to the serial position of a given item within the memory list. Recent behavioral, neuroimaging but also neuropsychological studies have shown that item and order retention capacities can be dissociated and depend, at least partly, on distinct mechanisms (see Attout, Van der Kaa, George & Majerus, 2012 and Majerus et al., 2010 for recent reviews; Poirier & Saint-Aubin, 1995; Henson et al., 2003; Nairne & Kelley, 2004; Lee & Estes, 1981). These studies have shown that storage of item information depends upon the richness of underlying phonological and semantic representations while order information storage reflects a specific, language-independent capacity. These dissociations are also in line with current theoretical models of verbal WM suggesting that item information is coded via temporary activation of the language knowledge base whereas order information is processed by distinct processes variously defined as temporal, contextual or spatial codes (Brown, Preece & Hulme, 2000; Henson, 1999; Page & Norris, 1998). Moreover, recent results go further and suggest that the order WM processes do not only support verbal WM tasks but also support other WM stimulus modalities; a number of studies have shown that order reconstruction during visual WM tasks leads to very similar serial position effects as in the verbal WM domain (Smyth, Hay, Hitch, & Horton, 2005) and that similar neural correlates (Majerus et al., 2007; 2010) underlie serial order processing in the verbal and visual WM domain. While some studies have focused on the role of order WM capacities in language development, no study has so far examined the role of order WM in numerical development. In the verbal domain, order WM, as opposed to item WM, has been shown to predict vocabulary learning (Gupta, 2003; Leclercq & Majerus, 2010) as well as decoding of novel words during the reading learning process (Martinez Perez, Majerus & Poncelet, 2012). These findings were interpreted as suggesting a role of order WM in the processing and learning of novel sequence information. This interpretation was recently supported by a study showing that the precision of serial

order coding in WM predicts the amount of phoneme position migration errors during a novel word learning task (Majerus & Boukebza, 2013). Likewise, in the numerical domain, retention and processing of sequence information is also likely to be a critical factor and this, particularly for efficient mental calculation. During mental calculation, not only the digits and operands have to be maintained, but critically, their sequential order has to be maintained ( $12 + 5 - 1$  will not lead to the same results as  $12 - 5 + 1$ ). Furthermore during early calculation development, the order of the numbers relative to their position in the number chain has to be co-activated in WM during the addition process: when adding 2 to 4 the child has to activate the fourth position of the number chain in WM, then go forward by two more positions and then stop when reaching the sixth position, update this position in WM and output the final result. These processes are highly sequential in nature and are likely to depend on well-developed sequential WM abilities. This is the main hypothesis that will be tested in this study.

A number of studies further attest of the importance of serial order processing in numerical processing and its association with WM. Ordinal processing abilities have been shown to underlie several aspects of numerical development such as ordinal knowledge of numbers (i.e., which number comes before/after another in the number sequence) (see Sury & Rubinsten, 2012 for a review). Rubinsten and Sury (2011) further investigated the representation of ordinality in adults with developmental dyscalculia and showed that adults with dyscalculia present specific difficulties to make numerical ordinal judgments whereas they presented a normal ratio-effect when processing magnitude. The authors suggested that ordinality and quantity are distinct systems while being both important for the development of numerical cognition (see also Lyons and Beilock, 2009, 2011). Similarly, Lyons and Beilock (2011) found that adults' performance in mental arithmetic was more strongly correlated with a numerical ordering task (judging if a triplet of Arabic digit is in ascending order) than with a

number magnitude task (selecting the larger of a pair of Arabic digit). A number of recent studies in healthy adults point to a number of similarities between ordinal processing in the numerical domain and WM, raising the possibility of shared cognitive processes between WM for order and numerical domains. Neuroimaging studies have shown that, the right anterior intraparietal sulcus is active both during coding of order information in WM tasks (Majerus et al., 2006a, 2010) and during ordinal judgment in numerical tasks (Kaufmann, Vogel, Starke, Kremser & Schocke, 2009; Turconi, Jemel, Rossion & Seron, 2004; Fias, Lammertyn, Caessens, & Orban, 2007). Likewise, some of the behavioral effects classically observed in numerical cognition such as the distance effect have also been shown to characterize WM performance (Moyer & Landauer, 1967). For example, Marshuetz, Smith, Jonides, DeGutis and Chenevert (2000) observed that the response times for determining whether two probe items are in the same serial order as in the memory list or not vary as a function of the positional distance of the two items within the memory list. A further hallmark effect of numerical cognition, the SNARC effect, has also been recently shown to appear in WM tasks by van Dijck and Fias (2011). The SNARC effect is characterized by faster response of the left hand to small numbers and faster response of the right hand to large numbers (Dehaene, Bossini & Giraux, 1993). Van Dijck and Fias observed that in WM tasks, items from the beginning of a memory list are responded to faster with the left hand and items from the end of the list are responded to faster by the right hand, reproducing, in the WM domain, the hallmark characteristics of the SNARC effect. In sum, the different studies discussed here suggest a close association between WM (measured in tasks that involve the retention of order) and ordinal processing in the numerical domain.

In light of these recent findings, we provide a new test of developmental associations between WM and calculation abilities. We test the existence of a longitudinal prediction of calculation abilities by WM abilities by focusing specifically on verbal WM for serial order,

given that the retention of order information is likely to be critical when maintaining and adding numbers and operands during mental calculation task, as discussed above. No study as so far has focused specifically on the developmental association between WM for serial order and calculation abilities, which may partly explain why no reliable associations have been documented so far between verbal short-term storage capacities and the development of calculation abilities. We will test this hypothesis by administering WM tasks maximizing storage capacities for serial order information as opposed to item information, and by administering numerical processing tasks (ordinal judgment, magnitude judgment, calculation) to children aged 5 years followed longitudinally over a three-year period.

At a secondary level, a supplementary hypothesis focuses more specifically on the reasons of this association by determining to what extent the association between order WM and calculation abilities may be mediated by access to a common set of ordinal representations, as suggested by some of the studies reviewed above. If this secondary hypothesis is true, then order WM and ordinal numerical processing should measure a common component, and hence they should predict each other in a longitudinal design, they should both predict calculation abilities, and the longitudinal prediction of later calculation abilities by order WM abilities at T1 should be mediated by ordinal numerical processing abilities at T1.

## **Study**

In order to test our hypotheses, a set of WM tasks and numerical processing tasks were administered to children at 3<sup>rd</sup> year of kindergarten (T1), at first (T2) and at second grade of primary school (T3), after onset of mathematical training. The WM tasks were adapted from previous studies exploring the relationship between WM development and language development (Leclercq & Majerus, 2010; Majerus et al., 2006b, 2009; Martinez Perez et al., 2012). WM for order information was assessed via a short-term serial order reconstruction

task which maximized serial order storage requirements, while minimizing phonological, lexical, and semantic information processing demands. In contrast to the serial order WM task, a single nonword delayed repetition task assessed WM for item information which maximized phonological processing demands by requiring the children to process, store, and repeat unfamiliar phonological information.

General mathematical abilities were assessed via a number addition task appropriate for kindergarten children with non-formal mathematical instruction (see also Noël, 2009) at T1 and T2. In this task the children had to add apples (from 1 to 8) and Arabic digits (from 1 to 3). At T3, mathematical abilities were assessed via a simple calculation task where children had to resolve a maximum of calculation operations in one minute and this for three types of operations (additions, subtractions and multiplications). We also tested the children's abilities to resolve more complex calculations by administering addition and subtraction tasks with carry over and multiplication tasks with operands higher than 3. Ordinal numerical abilities were assessed via an ordinal judgment task for numerical stimuli, adapted from previous studies (Jou & Aldrige, 1999; Turconi, Campbell & Seron, 2006; Gracia-Bafalluy & Noël, 2008). In this task, we presented two digits from 1 to 9 and the children were asked to judge whether the two digits were presented in canonical numerical order or not. Finally, a magnitude judgment task for numerical information was administered in order to assess the specificity of the potential associations between numerical ordinal judgment abilities and the WM and calculation measures. The numerical magnitude judgment task used the same stimuli and setup as the ordinal judgment tasks, but the children were instructed to decide which stimuli were larger rather than which stimuli came before another one.

## **Method**

### **Participants**



Sixty-nine children were tested at six months into third grade kindergarten (T1), one year later (T2), and again at second grade of primary school (T3), after having started formal arithmetic instruction. The children were recruited in twelve kindergarten and primary schools of the province of Liege, Belgium. All children came from families with middle-class socioeconomic background, as determined by their parents' professional status. For recruitment, parents were sent a written description of the study, an informed consent form, and an anamnestic questionnaire. This questionnaire allowed us to exclude children who did not speak French as a first language or who had had a history of neurological disorder, neurodevelopmental delay, sensory impairment or learning impairments. At T1, 68 kindergarten children (mean age: 68 months; range: 62–74; 33 girls and 35 boys) were retained for data inclusion; one participant had to be excluded as he presented a large proportion of abnormally low response times for the numerical ordinal and magnitude judgment tasks. At T2, we were able to retest 60 children of the initial group; three children had changed school, one no longer wished to participate, two children remained in kindergarten and two children had started speech therapy for written language disorders. At T3, we were able to retest 56 children of the initial group; one child had changed school, two children remained in first grade and one child had been advanced to grade 3.

## **Materials**

**Item WM task.** Item WM was assessed using a single nonword delayed repetition task already validated in several studies (Leclercq & Majerus, 2010; Majerus et al., 2006b, 2009; Martinez Perez et al., 2012). This task was designed to maximize the processing of phonological item information while minimizing the contribution of serial order WM processes. The task consisted of 30 monosyllabic nonwords presented in two different sessions allowing us to assess test-retest reliability. The stimuli had a CVC syllabic structure, and all were legal with respect to French phonotactic rules. This task minimized order

information retention requirements since only a single item had to be retained, all words had the same CVC structure and the items were recalled after a filled delay which hindered sequential rehearsal of the to-be-stored information. The nonword stimuli were recorded by a female human voice, stored on a computer disk and presented via headphones. At the end of each stimulus, the children were instructed to continuously repeat the syllable “bla” during 3 seconds. Then, the experimenter instructed the children to repeat the stimulus. The experimenter presented the task as follows in French [translate in English here] (see appendix A for original order in French): “You are an adventurer or a princess locked up in the tower of a castle. To find your way out of the castle, you have to open many doors by remembering passwords. When you see a closed door, you will hear through the headphones a password from a magic language and which you have to recall after having repeated ‘blablabla...’ during a short time. If you repeat correctly the password after my order, the door will open”. The children additionally had to repeat the nonword immediately after presentation to confirm that they had correctly perceived the item and were able to reproduce it accurately. However, no corrective feedback was given to the children. We determined the total number of phonemes correctly repeated after the delay as dependent variable, by pooling over the entire set of nonwords.

**Order WM task.** The retention of serial order information was assessed using a serial order reconstruction task also validated in several studies (Leclercq & Majerus, 2010; Majerus et al., 2006b, 2009; Martinez Perez et al., 2012). After the auditory presentation of sequences of animal names (*chat, chien, coq, lion, loup, ours, and singe* [cat, dog, cock, lion, wolf, bear, and monkey]), the children had to rearrange cards depicting the animals as a function of their order of presentation. This task minimized item information retention requirements since the seven stimuli were highly frequent (high lexical frequency, low age of acquisition and monosyllabic structure), known in advance and available at recall. The seven stimuli were

used to form lists with lengths ranging from two to seven items, and there were four trials for each list length (24 list trials). The sequences had been prerecorded by a female voice at a rate of one item per second, and were presented to the children via headphones. The trials were presented by increasing list length. At the end of each trial, the children were given cards (in alphabetical order) depicting the specific animals that had been presented in the trial. The children had to rearrange them according to their order of presentation, on a staircase with seven steps drawn on a sheet: they had to put the first item on the highest step, the second item on the second step, and so on. To explain the task to the child, the experimenter said in French [translate in English here] (see appendix A for original order in French): “Animals from all over the world gather to have a huge race. Seven animals participate: a cat, a dog, a cock, a lion, a wolf, a bear, and a monkey. Several races take place. Sometimes only two animals are participating. Sometimes there are three, four, five, six or seven animals. Through the headphones, you will hear someone announce the animal’s order of arrival at the finish line, from the first to the last animal. Immediately after, you have to put the pictures of the animals on the podium in their order of arrival. The animal arriving first has to be put on the highest step and the last one on the lowest step”. We determined the number of correctly placed items by pooling over all sequence lengths, as the dependent measure.

**Calculation.** Calculation abilities of the children were measured via simple addition tasks (see Noël, 2009 for a similar procedure). The children were asked to add two operands. The first operand was a drawing of a collection of items (apples) and the second was a simple Arabic digit (1, 2 and 3). The number of apples varied from 1 to 8 for additions with the digit 1, from 1 to 7 for additions with the digit 2 and from 1 to 5 for additions with the digit 3 in order not to exceed one-digit sums and arithmetic abilities of 5-to-6 year old children. The experiment was run on a PC using E-Prime 2.0 software. This task was presented visually and the child was asked to respond orally, the experimenter recording the child’s answer. The

entire test was comprised of 20 items, but since there was a time limit of 1 minute to perform the operations, the actual number of operations performed varied between participants; each child had nevertheless to perform a minimum of 10 additions. The problem was presented orally in French [translate in English here] (see appendix A for original order): “Look, here are three apples; if two more come, how many apples will there be?” We calculated the number of correct additions divided by the time taken to perform the task and then we multiplied this number by 60 in order to obtain the number of calculations performed in one minute (e.g. 10 additions in 80 seconds =  $(10/80)*60$ ). At T3, mathematical abilities were assessed via a simple calculation task testing simple automatized number facts (TTR, De Vos, 1992) with a paper-and-pencil test: Children had to solve a maximum of calculations in one minute and this for three types of operations (additions, subtractions and multiplications), by responding in writing. We also assessed more complex calculation abilities by administering addition and subtraction tasks with carry over and multiplication tasks for operands higher than 3. Presentation procedure and parameters were the same as for the calculations tasks presented at T1 and T2. The test was comprised of 18 items by type of operation to solve in 2 minutes. We calculated the number of correct operations for simple and complex calculation operations.

**Numerical ordinal and magnitude judgment tasks.** The ordinal and magnitude judgment tasks were performed with symbolic numbers (Arabic numerals). The tasks were run on a PC using E-Prime 2.0 software. Children were presented with two symbolic quantities disposed horizontally in one slide and were asked, in the ordinal condition, to decide whether the two stimuli were in canonical left-to-right order, and in the magnitude condition, to decide which one was the largest one. We used digits from 1 to 9 and the numerical distance between the two numbers varied between 1, 3 and 5 (we used the following pairs in ascending and descending order: for distance 1, the pairs were: 1-2, 2-3, 3-

4, 4-5, 5-6, 6-7, 7-8 and 8-9; for distance 3, the pairs were: 1-4, 2-5, 3-6, 4-7, 5-8 and 6-9; for distance 5, the pairs were: 1-6, 2-7, 3-8 and 4-9). Each condition (ordinal and magnitude judgments) contained 48 items with 16 items per distance. Participants were seated at 60 cm distance from the center of a 15'' computer screen. Each trial began with a fixation cross displayed for 2500 ms. After the disappearance of the fixation cross, the two stimuli appeared, one on the left side and the other on the right side of the screen and remained on the screen until the participant pressed one of the two response keys, with a response time limit of 7000 ms. For the ordinal condition, the response keys were 's' for 'no, there are not correctly ordered', and 'l', 'yes, they are correctly ordered'; for the magnitude condition, the response keys were 's' for 'the stimulus on the left side is the largest one', and 'l' for 'the stimulus on the right side is the largest one'. Both keys were marked with stickers depicting leftward or rightward pointing arrows for magnitude judgment and with smileys for ordinal judgment (angry face for no responses and smiling face for yes responses). Children were asked to make their decisions as quickly and as accurately as possible. For the ordinal conditions, the experimenter said in French [translate in English here] (see appendix A for original order) the following while pointing on the corresponding locations of the screen: "We asked another child to arrange from the first to the last two numbers. Sometimes he was wrong. Can you indicate him if he correctly ordered the numbers in pressing the key 'yes' or 'no'?". For the magnitude condition, the experimenter said the following: "Imagine that the numbers represent the quantity of candies that you might have, you have to choose the side where there is the most in pressing the key on the same side". For each condition, a block of 3 practice trials was presented first, followed by the experimental trials with a total of 48 trials including 16 trials per distance. The median reaction time for correct responses was chosen as the dependent variable in the magnitude and ordinal judgment tasks (Turconi et al., 2006; Franklin, Jonides & Smith, 2009). In order to control for general processing speed shared by

both measures, each correlation between median reaction times of the ordinal judgment task and another variable was partialized by median reaction times of the magnitude judgment task, and vice-versa.

**Nonverbal intelligence and vocabulary knowledge.** We also collected estimates of nonverbal intelligence and vocabulary knowledge to control for overall nonverbal and verbal abilities in our participants. Raven's Colored Progressive Matrices (RCPM; Raven, Court & Raven, 1998) were used as a measure of general nonverbal reasoning abilities. The raw scores were taken as the dependent measure. Receptive vocabulary knowledge was estimated using the EVIP scales (*Echelle de Vocabulaire en Images Peabody*) (Dunn, Thériault & Dunn, 1993). As a dependent variable, we used the raw vocabulary scores.

### **Procedure**

All tests were administered at each time. The tasks were administered in three sessions at T1 and two sessions at T2 and T3. In order to avoid fatigue effects, each experimental task was administered in two parts, with half of the trials administered for each part. At T1, the children were first administered a digit reading task (digits from 1 to 10) in order to exclude children who did not have sufficient digit knowledge to perform the tasks; digit knowledge was assessed by asking the children to name all digits from 1 to 10 printed on an A4 sheet. The experimental tasks were the following order: In the first session, we administered the first halves of the item WM, order WM, ordinal judgment and the magnitude judgment tasks. The second session (1 week later) involved the administration of the Raven's matrices and vocabulary knowledge (EVIP) and calculation tasks. In the third session we administered the second halves of the item WM, order WM, ordinal judgment and the magnitude judgment tasks. At T2 and T3, on the first session, we administered the first halves of the item WM, order WM, ordinal judgment and the magnitude judgment tasks but also the Raven's matrices. In the second session we administered the second halves of the item WM, order WM, ordinal

judgment and the magnitude judgment tasks but also vocabulary knowledge (EVIP) and calculation tasks. Children were tested individually in their respective school, in a quiet room. Statistical analyses were performed using STATISTICA 9 software package (StatSoft, Tulsa, Oklahoma, USA).

## Results

Valid data sets were obtained for 68 participants at T1, for 60 participants at T2 and for 56 at T3. Statistics comparing data obtained at T1 and T2 were conducted on the same subset of 60 participants and statistics comparing data obtained at T1, T2 and T3 included 56 participants. Descriptive statistics for all variables are presented in Table 1. The distribution of scores for the different tasks was assessed by determining skewness and kurtosis (Tabachnick & Fidell, 1996) (see Table 1). The vast majority of measures had acceptable skewness and kurtosis (values within the recommended two standard errors range). This was however not the case for accuracy on the ordinal and magnitude judgment tasks due to a ceiling effect (see Table 1); we therefore chose to use median response times as the dependent measure on this task (see also Methods section) since skewness and kurtosis values for median response times remained in the recommended range, except for minor deviations for a few time points. For the addition task at T2, skewness exceeded the cut-off value for a very minimal amount (.03); kurtosis was well below cut-off. For the item WM task at T2 and T3, skewness exceeded the cut-off value for a very minimal amount (respectively .03 and .20); kurtosis was well below cut-off.

The content validity of ordinal judgment and magnitude judgment tasks was confirmed by a different pattern of distance effects at each time point (see Table S1 in Supplemental Material), as already observed in previous studies with adults (Turconi et al., 2006; Franklin et al., 2009; Lyon & Beilock, 2013; 2011; Rubinsten et al., 2011; 2013) and

children (Colomé & Noël, 2012; Kaufmann et al., 2009). This was confirmed by a 2 (Task) x 3 (Distance) x 2 (Order) ANOVA on response times for each time point leading to the expected three-way interaction (T1:  $F_{(2,114)}=7.32$ ,  $\eta^2 = 0.11$ ,  $p<.01$ ; T2:  $F_{(2,116)}=7.26$ ,  $\eta^2 = 0.11$ ,  $p<.01$ ;  $F_{(2,110)}=10.30$ ,  $\eta^2 = 0.16$ ,  $p<.001$ ). For the WM tasks, we checked that the item WM task minimized serial order WM processes, by computing the number of order errors (within-stimulus phoneme migrations) in this task. As predicted, these inversions were very rare, only 2.33 % errors being due to inversions of the first and last consonant. Finally, we computed reliability coefficients (Cronbach's alpha) for all tasks except for already standardized tests (EVIP, Raven). Reliability coefficients for WM (.65 to .87), calculation (.73 to .91) and judgment tasks (.94 to .97, for RTs) were in a satisfactory to excellent range.

< INSERT TABLE 1 ABOUT HERE >

### Cross sectional analyses

A first set of analyses determined the associations between the WM and numerical processing tasks separately for each time point of assessment. We computed partial correlations between order WM, item WM, and numerical processing tasks after control for age (in months), nonverbal intelligence and vocabulary knowledge at each time. At T1, as shown in Table 2, the order WM task but not the item WM task correlated significantly with the addition task (respectively,  $CI_{95\%} = .063$  to  $.513$  and  $CI_{95\%} = -.044$  to  $.399$ ). A multiple regression analysis determined whether order WM still predicts addition abilities after controlling for general WM processes shared with item WM (see Table S2 in Supplemental Material). Age, nonverbal intellectual efficiency, vocabulary knowledge and item WM scores were entered in the regression at steps 1, 2, 3 and 4, respectively, followed by the order WM score. This analysis revealed that order WM explained significantly 5% of unique variance in addition abilities ( $\beta = .25$ ,  $t_{(62)}=2.08$ ,  $p<.05$ ). At T2, both item and order WM tasks were found



to correlate significantly with addition abilities (see Table 2) (order WM:  $CI_{95\%} = .242$  to  $.601$ ; item WM:  $CI_{95\%} = .096$  to  $.586$ ). A multiple regression analysis determined whether order or item WM still predicts addition abilities after controlling for age, nonverbal intellectual efficiency, vocabulary knowledge and item WM or order WM scores. This analysis revealed that order WM explained 8% of unique variance in addition abilities ( $\beta = .33$ ,  $t_{(54)}=2.58$ ,  $p<.05$ ) while item WM did not significantly explain unique variance (4%) in addition abilities ( $\beta = .23$ ,  $t_{(54)}=1.79$ ,  $p=.08$ ) (see Table S2 in Supplemental Material). Finally, at T3, both item and order WM tasks were found to correlate with simple calculation abilities, nearly significant for item WM, ( $p=.059$ ), and just significant for order WM ( $p=.049$ ) (order WM:  $CI_{95\%} = -.038$  to  $.542$ ; item WM:  $CI_{95\%} = -.055$  to  $.536$ ) but only the order WM task correlated significantly with the complex calculation task (see Table 2) (order WM:  $CI_{95\%} = .223$  to  $.668$ ; item WM:  $CI_{95\%} = -.139$  to  $.516$ ). Multiple regression analyses revealed that order WM explained 16% of independent variance of complex calculation ( $\beta = .48$ ,  $t_{(50)}=3.41$ ,  $p<.01$ ) but not a unique variance of simple calculation (4%) ( $\beta = .23$ ,  $t_{(50)}=1.59$ ,  $p=.12$ ) (see Table S2 in Supplemental Material). At each time point, correlations between numerical magnitude judgment and calculation abilities were non-significant while numerical ordinal judgment and calculation abilities correlated significantly at T2 and T3. Moreover, the two judgment tasks correlated at each time, which is not surprising given that both tasks involve a measure of processing speed.

< INSERT TABLE 2 ABOUT HERE >

### **Longitudinal analyses**

Next, we assessed the longitudinal association between calculation and WM abilities. In order to do this, we correlated WM abilities at T1 with calculation abilities at T2 and T3, while

controlling each time for age, vocabulary and non-verbal intelligence at T1. When correlating WM abilities at T1 with addition abilities at T2, we observed a significant association only between addition abilities and the order WM measure (order WM:  $CI_{95\%} = .066$  to  $.527$ ; item WM:  $CI_{95\%} = -.181$  to  $.409$ ) (see Table 2 and Fig. 1). In order to assess the specificity of this association, a multiple regression analysis determined whether order WM at T1 still predicts addition abilities at T2 after controlling for general WM processes shared with item WM at T1 (see Table S3 in Supplemental Material). Age, nonverbal intellectual efficiency, vocabulary knowledge and item WM scores were entered in the regression at steps 1, 2, 3 and 4, respectively, followed by the order WM score. This analysis revealed that order WM explained 7% of unique variance in first grade mathematics achievement ( $\beta = .31$ ,  $t_{(54)} = 2.14$ ,  $p < .05$ ).

When correlating WM abilities at T1 with calculation abilities at T3, after controlling for age, vocabulary and non-verbal intelligence at T1, we observed a marginally significant correlation between item WM at T1 and simple calculation at T3 ( $r = .26$ ,  $p = .058$ ) (order WM:  $CI_{95\%} = -.116$  to  $.434$ ; item WM:  $CI_{95\%} = -.033$  to  $.542$ ); since simple calculation rely on verbal routines and retrieval of these routines in long-term memory, this correlation suggests a link between long-term memory retrieval of arithmetic facts and temporary activation of phonological long term memory which is considered to be measured by the item WM task (Gathercole, Frankish, Pickering & Peaker, 1999; Majerus, Van der Linden, Mulder, Meulemans & Peters, 2004). Moreover, we observed a significant correlation between order WM at T1 and complex calculation at T3 (order WM:  $CI_{95\%} = .050$  to  $.536$ ; item WM:  $CI_{95\%} = -.170$  to  $.444$ ). A multiple regression analysis revealed that order WM explained 8% of unique variance of complex calculation ( $\beta = .34$ ,  $t_{(50)} = 2.13$ ,  $p < .05$ ) (see Table S3 in Supplemental Material).

On the other hand, we observed that WM capacities at T1 correlated significantly with WM abilities at T2 but not at T3, while WM capacities at T2 correlated significantly with those at T3. The reduced correlation between order WM measures at T1 and T3 is likely to be due to additional variables affecting order WM performance at T3. Subvocal articulatory rehearsal might be a good candidate to explain the developmental change between T1 and T3 since we know that children younger than 7 years do not use rehearsal strategies spontaneously, or do so only inefficiently (Gathercole, Adams, & Hitch, 1994; Cowan et al., 1994; Cowan & Kail, 1996). Hence, it is likely that individual differences in articulatory rehearsal did not strongly influence order WM performance at T1, started to influence order WM performance inconsistently at T2, and this influence became more systematic at T3, which may explain the reduction of the correlation strength between order WM at T1 and T3 (note that the correlation although non-significant at the  $p < .05$  level, was nevertheless close to significance,  $p = 0.076$ ). In order to provide further support for the developmental associations between order WM abilities and calculation abilities, we checked whether the relation between order WM at T1 and calculation at T3 was mediated by the calculation abilities at T1, in which case the impact of order WM would only be a very short-lived one. Multiple regression analyses revealed that order WM at T1 still explained 8% of unique variance of complex calculation ( $\beta = .31$ ,  $t_{(50)} = 2.22$ ,  $p < .05$ ) when entering calculation abilities at T1 in the regression at step 4. Furthermore, when entering order WM abilities at T3 in the regression at step 4, order WM at T1 did not explain a unique variance of complex calculation anymore ( $\beta = .19$ ,  $t_{(50)} = 1.63$ ,  $p > .05$ ), suggesting that the same processes may explain the association for order WM at T1 and calculation abilities at T3 and relative to the association of order WM at T3 and calculation abilities at T3 (see Table S2 in Supplemental Material). At the same time, at T3, order WM still explained calculation abilities when entering order WM abilities at T1 in the regression at step 4 ( $\beta = .50$ ,  $t_{(50)} = 4.12$ ,  $p > .001$ ), indicating that rehearsal strategies

likely to intervene more efficiently at T3, also possibly play a role in calculation abilities (see Table S2 in Supplemental Material). This assumption of additional implication of rehearsal processes could also explain why the association between order WM and calculation abilities was particularly strong at this time point. Overall, these results indicate that the development of calculation abilities is supported by order WM processes at different developmental stages, with a potentially increasing impact of sequential rehearsal processes at later developmental stage.

Finally, when correlating WM abilities at T2 with calculation abilities at T3, after controlling for age, vocabulary and non-verbal intelligence at T2, we only observed a significant correlation between item WM at T2 and simple calculation at T3 ( $CI_{95\%} = -.001$  to  $.537$ ). This correlation did not remain significant when controlling also for order WM abilities at T2 ( $r = .22$ ,  $p > .05$ ). The absence of correlation between order WM at T2 and calculation abilities at T3 could again be related to the emerging and inconsistent influence of rehearsal strategies in the order WM measure at T2 which will add an additional source of variability to task performance.

In the same way, we assessed the longitudinal association between numerical judgment abilities and calculation. Magnitude judgment abilities at T1 correlated significantly with addition abilities at T2 and with simple and complex calculation at T3 in line with previous studies (Holloway & Ansari, 2009; De Smedt, Verschaffel, & Ghesquière, 2009; Bugden, Price, McLean & Ansari, 2012). On the other hand, ordinal judgment abilities did not correlate significantly with addition abilities one year and two years later (after control of magnitude judgment abilities). The fact that ordinal abilities correlate with calculation abilities at a cross-sectional level while magnitude judgment abilities at T1 predict calculation abilities one and two years later suggests that ordinal and magnitude processing abilities have different roles in the development of numerical abilities. Further, we observed that ordinal

judgment capacities at T1 correlated with the same measure at T2, and the measure at T2 correlated with the measure at T3, This indicates a progressive change in processes underlying this task, with the turning point being at T2, when children were six months into 1<sup>st</sup> grade and thus had started formal mathematics training. A possible interpretation it is that formal, explicit mathematics training will enrich and detail numerical ordinal representations, and especially will automatize access to ordinal representations via repeated training and use of these representations. Previous studies showed that ordinal meaning of numbers is mastered later than the cardinal meaning (Colomé & Noël, 2012; Miller, Major, Shu, & Zhang, 2000). In five-year old children (roughly the same age as those tested at T1 in this study) performance in a “give-me” task with cardinal meaning (i.e., determine the number of objects) (90%) is significantly higher as compared to a “give-me” task with ordinal meaning (i.e., determine the object shown in a given numerical position) (63%) (Colomé & Noël, 2012), suggesting that in five-year olds, the ordinal numerical abilities are far from being mastered. This is likely to explain why associations between calculation abilities and ordinal abilities were not significant at T1, but became significant at T2 and T3. We also observed that item WM abilities at T1 showed a significant correlation with magnitude judgment abilities at T3 but not at T2, urging for caution in the consideration of this correlation (see Table 2).

Overall, the results confirm our main hypotheses assuming a more important implication of early order WM processing in the development of calculation abilities by showing that early order processing in WM predicts calculation abilities, one year and two years later, and this independently of item WM capacities. Moreover, the item WM abilities appear to be more related to arithmetic fact retrieval. On the other hand, the results confirm previous studies showing an association between early magnitude processing abilities and

later calculation abilities and by showing a correlation at cross-sectional level between numerical ordinal abilities and mathematical achievement.

< INSERT FIGURE 1 ABOUT THERE >

Finally, we explored the link between order WM, calculation abilities and numerical ordinal processing abilities in order to better understand the nature of the association between order WM abilities and calculation abilities. There was no longitudinal correlation between early ordinal judgment abilities and later order WM abilities (see Table 2). We also checked whether the developmental changes in ordinal abilities may have influenced the associations observed between order WM and calculation abilities. When controlling for ordinal abilities at either T1 or T3, the relation between order WM at T1 and calculation abilities at T3 remained significant (respectively,  $r=.31$ ,  $p<.05$ ;  $r=.31$ ,  $p<.05$ ). These results do not support the hypothesis that the association between order WM and calculation abilities would be mediated by access to numerical ordinal representations; rather they suggest that order WM and ordinal numerical processing do not measure a common component at the particular developmental stages investigated here and for the particular tasks used here.

### **Discussion**

This longitudinal study used the distinction between item and order information in WM in order to achieve a better understanding of the relationship between early verbal WM and later numerical development. We observed that serial order WM capacities, as opposed to item WM capacities, are the most reliable predictor of later calculation abilities in 6-to-8 year-old children, even when general cognitive efficiency is controlled for. We also observed that although early serial order WM capacities and numerical ordinal judgment abilities are both

linked to calculation abilities, the association between order WM and calculation abilities is not mediated by access to numerical ordinal representations.

### **The link between WM and calculation abilities**

Past studies have shown inconsistent associations between verbal WM and calculation abilities (Hecht et al., 2001; Swanson & Kim, 2007; Noël et al., 2004; Noël, 2009; De Smedt et al., 2009). The distinction between item and order information in verbal WM allowed us in the present study to demonstrate a robust and a stronger link between calculation abilities and order WM capacities. Previous studies generally used multi-determined WM tasks confounding order and item levels of processing, such as multisyllabic nonword repetition tasks or word list immediate serial recall tasks (Passolunghi et al., 2007; Gathercole & Pickering, 2000b; De Smedt et al., 2009). These tasks require to simultaneously store the items and their order of presentation when repeating a list of nonword syllables or words. The present study shows that, by using verbal WM tasks more specifically designed to measure those WM components that may be critical for numerical development, a more reliable association between verbal WM measures and numerical development can be obtained. We indeed observed that order WM abilities are a more robust independent predictor of calculation abilities one year later and two years later than item WM. This was confirmed by multiple regression analyses showing that order WM predicted calculation abilities after controlling for more general WM processes shared with the item WM task. Item WM predicted merely simple multiplication two years later, and this prediction was not specific since it was not significant anymore after control of the contribution of order WM abilities. Moreover, at each time point, order WM predicted calculation abilities and this independently of item WM, as confirmed by multiple regression analyses. At the same time, it should be noted that performance on item and order WM tasks is not completely independent, since recruitment of attentional and executive control is a further important determinant of WM

capacities (Cowan, 1995; Majerus et al., 2009), and of calculation abilities more generally (LeFevre et al., 2013); this is also supported by the significant inter-correlations between the two WM tasks. However, despite these processes shared between order and item WM tasks, only the order WM measure emerged as a robust and independent predictor of later calculation abilities, indicating that the processes more specifically recruited by the order WM task, namely the ability to maintain but also to rehearse serial order information, is an important building block of arithmetic development.

### **The link between numerical ordinal processing and calculation abilities**

Most previous studies on determinants of calculation abilities focused on numerical magnitude rather than ordinal processing, and showed that numerical magnitude judgment correlates with calculation abilities in children at first grade of primary school (Holloway & Ansari, 2009; De Smedt et al., 2009; Bugden et al., 2012). In line with these studies, the present study observed an association between early numerical magnitude processing and mathematical development, by showing that the median RTs of magnitude judgment (after control of ordinal judgment) at T1 predicted calculation abilities one and two years later.

A novel finding of this study is to highlight an association between ordinal levels of representation and calculation abilities. Few studies have studied the link between numerical ordinal processing and calculation abilities (Lyons & Beilock, 2011; Rubinsten & Sury, 2011) and none at a longitudinal level. Ordinal processing has been considered recently as being as important as is quantity processing for predicting numerical abilities in young children (Rubinsten & Sury, 2011; Kaufmann et al., 2009; Kucian, Loenneker, Martin & von Aster, 2011) and are also consistent with the recent model of Sury and Rubinsten (2012) assuming that ordinality is a crucial building block of the development of numerical representations. In the present study, we observed a strong link between numerical ordinal abilities and calculation abilities but only at a cross-sectional level and not at a longitudinal level unlike the



results observed for the numerical magnitude abilities. These data thus do not support a causal association between numerical ordinal processing abilities and later calculation abilities, but they suggest, at the least, that ordinal processing and calculation abilities are associated, and this via different pathways than magnitude processing abilities. According to Lyons and colleagues, ordinal representations for numerical symbols serve as an intermediary step that links the magnitude representation and more complex arithmetic abilities (Lyons & Ansari, 2009; Lyons & Beilock, 2009). Ordinal abilities would refine the representation of numbers, with processing increasingly based on numerical symbols, leading to a more fine-grained mental number line. This hypothesis is supported by recent studies indicating that the acquisition of ordinal and cardinal meanings of numbers are independent, with ordinality being mastered after the development of cardinality (Colomé & Noël, 2012). This is also consistent with our data showing that only from T2 onwards, numerical ordinal judgment abilities correlated with calculation abilities.

### **Distinct ordinal processes linked to mathematical development**

With respect to the association of order WM and numerical cognition, we tried to determine more specifically the nature of this association by exploring the role of access to numerical ordinal representations in this association. We observed that there was no longitudinal correlation between ordinal judgment abilities and order WM capacities across time points nor at a cross-sectional level. Moreover, we observed that the associations between order WM and calculation remained significant when controlling for ordinal abilities at either T1 or T3. Our data therefore show that order WM capacities predict calculation abilities independently of numerical ordinal abilities. This indicates that the order WM abilities predict calculation abilities not via access to a common set of (long-term) ordinal representations but rather via mechanisms intrinsically associated with short-term storage capacities of order information.

As regards the association between order WM and calculation abilities, order WM abilities involve serial order storage processes but could also involve serial rehearsal processes which are likely to be recruited by the order WM task at later developmental stages and which further contribute to the maintenance of serial order information (Gupta & MacWhinney, 1997; Henson et al., 2003). These results further our understanding of the relationship between verbal WM and numerical development, by showing that the early processes involved in short-term maintenance of serial order information are critical on their own for performing calculation tasks and for predicting later mental calculation abilities while the later appearing sequential rehearsal strategies could be an additional factor supporting calculation abilities. As already noted in the Introduction, this is very likely related to the strong sequential processing demands of mental calculation tasks: in these tasks, the maintenance and rehearsal of numbers and operands in correct serial position is critical for accurate calculation results, especially when updating intermediate calculation results during complex calculation (Noël et al., 2004). The later appearing rehearsal processes may be particularly important for maintenance and updating of intermediate representations involved in carrying and borrowing procedures during complex calculations. Furthermore, during early calculation development, WM is also needed to co-activate the order of the numbers relative to their position in the number chain and to use and update this sequential information in WM, which will allow accurate computations by scanning the positions of the mental number line in a forward or backward manner for the number of positions required by the computation. This operation strongly relies on sequential activation and rehearsal of the positions of the mental number line in WM. This interpretation also predicts that with progressive automation and proceduralisation of arithmetic abilities, the link between WM for order and mental calculation may decline, and only reappear for more complex calculations, as we did observe in this study.

To conclude, the present study shows a clear association between calculation abilities and the short-term maintenance of serial order information in WM. Future studies will need to determine the full range of arithmetic operations that depend upon serial order WM capacities, the present study providing a first attempt by demonstrating a link between serial order WM and early addition, subtraction and multiplication abilities.

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Table 1

*Descriptive statistics for variables at T1, T2 and T3.*

|    | Measures                       | M      | SD    | Range     | Skewness | Kurtosis |
|----|--------------------------------|--------|-------|-----------|----------|----------|
| T1 | Age                            | 67.8   | 3.4   | 62-74     | 0.15     | -1       |
|    | EVIP                           | 73.9   | 14.2  | 49-111    | 0.49     | -0.16    |
|    | RCPM                           | 20.8   | 3.7   | 13-29     | 0.32     | -0.32    |
|    | Addition                       | 6.4    | 3.11  | 0-13      | 0.06     | -0.57    |
|    | Item WM                        | 57.8   | 13.4  | 16-83     | -0.48    | 0.17     |
|    | Order WM                       | 47.6   | 8.8   | 30-70     | 0.19     | -0.37    |
|    | Magnitude judgment (ACC)       | 0.91   | 0.06  | 0.68-0.98 | -1.55*   | 2.50*    |
|    | Magnitude judgment (median RT) | 2343.7 | 604.1 | 1389-4265 | 0.89*    | 0.66     |
|    | Ordinal judgment (ACC)         | 0.80   | 0.14  | 0.5-0.98  | -0.59*   | -0.99    |
|    | Ordinal judgment (median RT)   | 3700   | 803.3 | 2041-5531 | 0.22     | -0.49    |
| T2 | Age                            | 79.9   | 3.4   | 74-86     | 0.15     | -1.08    |
|    | EVIP                           | 86.3   | 14.7  | 49-117    | -0.60    | 0.33     |
|    | RCPM                           | 23.6   | 4     | 13-34     | -0.03    | 0.36     |
|    | Addition                       | 12.3   | 3     | 5.7-18    | -0.65*   | 0.05     |
|    | Item WM                        | 70.3   | 8.8   | 45-84     | -0.65*   | 0.20     |
|    | Order WM                       | 58     | 8.6   | 39-75     | -0.48    | -0.40    |
|    | Magnitude judgment (ACC)       | 0.95   | 0.03  | 0.8-1     | -2.07*   | 6.3*     |
|    | Magnitude judgment (median RT) | 1633.9 | 447.4 | 897-3016  | 0.82*    | 1.15     |
|    | Ordinal judgment (ACC)         | 0.92   | 0.07  | 0.5-1     | -3.88*   | 21.8*    |
|    | Ordinal judgment (median RT)   | 2781.9 | 650   | 1541-4222 | 0.25     | -0.33    |
| T3 | Age                            | 92.1   | 3.4   | 86-98     | 0.08     | -1.09    |
|    | EVIP                           | 97.7   | 13.1  | 72-135    | 0.36     | 0.59     |
|    | RCPM                           | 27.3   | 3.2   | 20-34     | 0.14     | -0.13    |
|    | Simple calculation             | 31.8   | 7.8   | 12-47     | -0.27    | 0.17     |
|    | Complex calculation            | 25     | 8.2   | 7-47      | 0.12     | 0.57     |
|    | Item WM                        | 79.6   | 5.6   | 63-88     | -0.84*   | 0.60     |
|    | Order WM                       | 69.8   | 8.8   | 44-88     | -0.21    | 0.31     |
|    | Magnitude judgment (ACC)       | 0.95   | 0.03  | 0.9-1     | -0.86*   | 1.44*    |
|    | Magnitude judgment (median RT) | 1071.9 | 277.8 | 647-2043  | 1.06*    | 1.38     |
|    | Ordinal judgment (ACC)         | 0.92   | 0.05  | 0.8-1     | -1.72*   | 3.07*    |
|    | Ordinal judgment (median RT)   | 1811.1 | 584.9 | 688-3768  | 0.80*    | 1.08     |

*Note.* RCPM = Raven's Colored Progressive Matrices; EVIP = French version of Peabody Picture Vocabulary Test; WM = Working Memory; ACC=Accuracy; RT = Response Time.

\*= exceeds level of 2 standard error skewness cutoff = 0.58 for T1, 0.62 for T2, 0.64 for T3 and kurtosis cutoff = 1.15 for T1, 1.22 for T2, 1.26 for T3.

## ORDER WM AND NUMERICAL DEVELOPMENT

Table 2

*Cross-sectional partial correlations between tasks for T1, T2 and T3 and longitudinal partial correlations (in grey) between T1 and T2, T1 and T3 as well as between T2 and T3; variables partialled out: age, vocabulary knowledge and nonverbal intelligence.*

|                                    | T1 |       |      |      |        | T2     |                       |        |        |                        | T3     |                       |                       |                        |       |                       |
|------------------------------------|----|-------|------|------|--------|--------|-----------------------|--------|--------|------------------------|--------|-----------------------|-----------------------|------------------------|-------|-----------------------|
|                                    | 1  | 2     | 3    | 4    | 5      | 6      | 7                     | 8      | 9      | 10                     | 11     | 12                    | 13                    | 14                     | 15    | 16                    |
| 1. Item WM                         | /  | .36** | .17  | -.08 | .07    | .57*** | .20                   | .14    | .05    | -.19                   | .16    | .04                   | .26 <sup>(.058)</sup> | .14                    | -.21  | .40**                 |
| 2. Order WM                        |    | /     | .30* | .00  | -.03   | .45*** | .58***                | .31*   | .00    | -.10                   | .10    | .25                   | .19                   | .32*                   | -.14  | .13                   |
| T1 3. Addition                     |    |       | /    | -.02 | -.22   | .22    | .25 <sup>(.062)</sup> | .51*** | -.12   | -.02                   | -.10   | .05                   | .33*                  | .11                    | .04   | -.08                  |
| 4. Ordinal judgment (median RT)    |    |       |      | /    | .49*** | -.20   | -.09                  | -.03   | .27*   | -.25 <sup>(.068)</sup> | -.06   | .04                   | .15                   | .13                    | .06   | .01                   |
| 5. Magnitude judgment (median RT)  |    |       |      |      | /      | -.16   | -.00                  | -.33*  | .02    | .30*                   | -.14   | -.12                  | -.47***               | -.40**                 | .09   | .26 <sup>(.061)</sup> |
| 6. Item WM                         |    |       |      |      |        | /      | .41**                 | .37**  | -.13   | -.01                   | .50*** | .05                   | .30*                  | .24                    | -.16  | .03                   |
| 7. Order WM                        |    |       |      |      |        |        | /                     | .43**  | -.13   | .12                    | .17    | .30*                  | .24                   | .21                    | -.06  | -.24                  |
| T2 8. Addition                     |    |       |      |      |        |        |                       | /      | -.35** | -.05                   | .18    | .23                   | .48***                | .40**                  | -.11  | -.31*                 |
| 9. Ordinal judgment (median RT)    |    |       |      |      |        |        |                       |        | /      | .65***                 | .01    | -.10                  | -.20                  | -.07                   | .36** | .07                   |
| 10. Magnitude judgment (median RT) |    |       |      |      |        |        |                       |        |        | /                      | -.09   | .01                   | -.17                  | -.25 <sup>(.069)</sup> | .07   | -.05                  |
| 11. Item WM                        |    |       |      |      |        |        |                       |        |        |                        | /      | .26 <sup>(.063)</sup> | .26 <sup>(.059)</sup> | .22                    | -.14  | -.08                  |
| 12. Order WM                       |    |       |      |      |        |        |                       |        |        |                        |        | /                     | .27*                  | .47***                 | -.24  | -.06                  |
| T3 13. Simple calculation          |    |       |      |      |        |        |                       |        |        |                        |        |                       | /                     | .66***                 | -.24  | -.24                  |
| 14. Complex calculation            |    |       |      |      |        |        |                       |        |        |                        |        |                       |                       | /                      | -.28* | .01                   |
| 15. Ordinal judgment (median RT)   |    |       |      |      |        |        |                       |        |        |                        |        |                       |                       |                        | /     | .55***                |
| 16. Magnitude judgment (median RT) |    |       |      |      |        |        |                       |        |        |                        |        |                       |                       |                        |       | /                     |

*Note.* WM = Working Memory; RT = Response Times. \* $p < .05$ ; \*\* $p < .01$ ; \*\*\*  $p < .001$ .

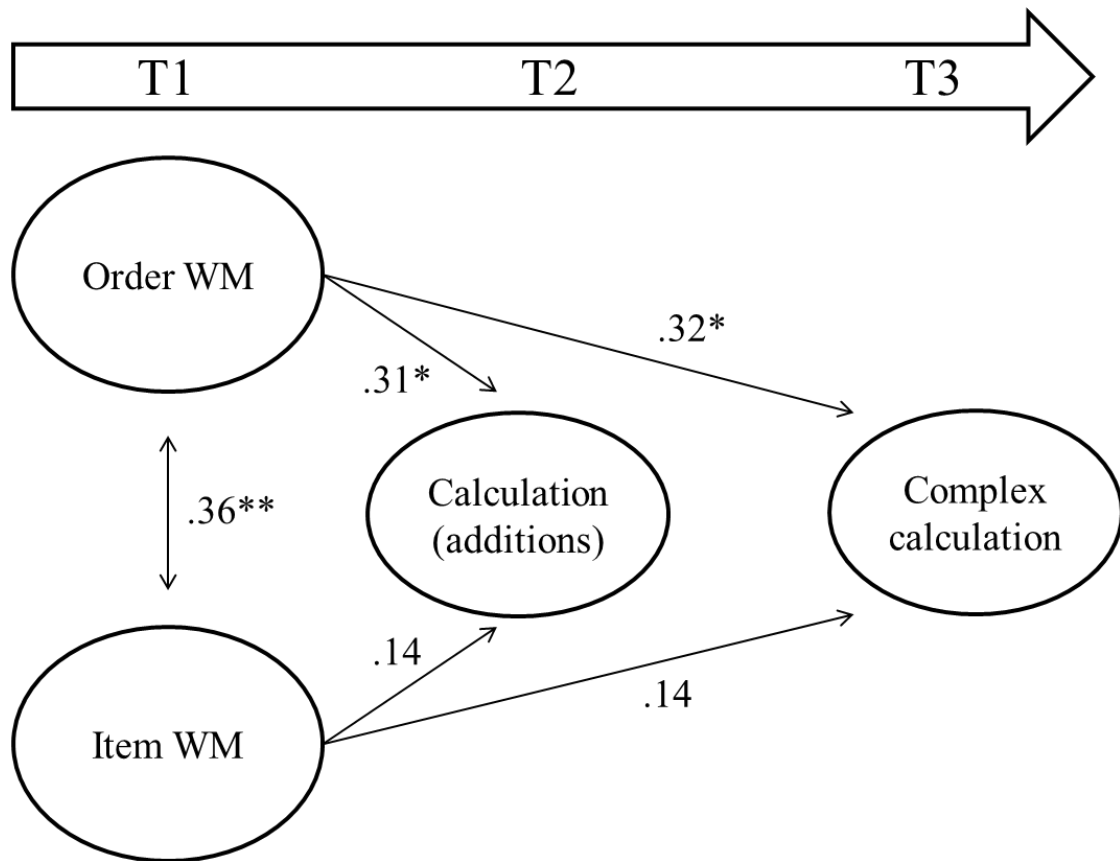


Figure 1. Illustration of partial correlations between WM abilities at T1 and calculation abilities at T2 and T3 with age, vocabulary knowledge and nonverbal intelligence partialled out. WM = Working memory.  $*p < .05$ ;  $**p < .01$ .



## APPENDIX A

*Original order in French to different tasks*

Addition task: « *Tu vas voir apparaître des calculs à l'écran. Additionne les pommes avec le nombre qui le suit et dis-moi la réponse tout haut. Résous le plus possible de calcul correctement.* »

Ordinal judgment task: « *Nous avons demandé à un autre enfant de ranger les deux nombres du premier au dernier. Parfois il a commis des erreurs. Peux-tu indiquer si il les a rangés dans le bon ordre en appuyant sur la touche oui ou non ?* »

Magnitude judgment task: « *Imagine que les nombres représentent une quantité de bonbons que tu aurais, peux-tu choisir le côté où il y en a le plus en appuyant sur la touche du même côté.* »

Order WM: « *Chaque année, des animaux du monde entier se réunissent pour participer à une grande course à pied. Cette année, sept animaux participent à la course: un chien, un chat, un lion, un ours, un loup, un singe et un coq. Plusieurs courses sont organisées. Parfois, seulement deux animaux participeront. Parfois, il y en aura trois, quatre ou cinq. Et d'autres fois, il y aura des grandes courses auxquelles six ou sept animaux participeront. Je vais te dire l'ordre dans lequel les animaux ont passé la ligne d'arrivée, du premier au dernier. Il faudra ensuite que tu mettes les dessins de ces animaux sur le podium d'après l'ordre dans lequel ils sont arrivés. Le premier devra être placé sur la plus haute marche du podium et le dernier sur la plus basse. Tu as bien compris?* »

Item WM: « *Tu es un aventurier/une princesse enrôlé(e) dans la tour d'un château. Ce château est rempli de portes que tu dois ouvrir pour sortir. Chaque porte s'ouvre grâce à un mot de passe que tu vas entendre dans le casque. Mais attention, le mot de passe est en langue magique ! Tu devras bien le retenir. La porte s'ouvrira si tu répètes tout de suite le mot de passe après l'avoir entendu, puis que tu dis « blabla... » jusqu'à ce que je te dise « le mot de passe est ... » et alors tu redis le mot de passe. Si tu ne connais plus bien le mot de passe tu as 3 clés à ta disposition pour ouvrir les portes. Tu as bien compris ?* »